

Problem #1: Number Analyzer

A Step-by-Step Console Application Tutorial

Step 1

Create New Project

Setting up your Console Application in Visual Studio.



Project Setup

- 1 Launch **Visual Studio**
- 2 Click "**Create a new project**"
- 3 Search for "**Console App (.NET Framework)**"
- 4 Select the **C# version**



Configuration

Project Name:

NumberAnalyzerConsole

Location:

Choose your folder

Framework:

.NET Framework

Click 'Create' to generate your project!

Step 2

Understand the Core Logic

Familiarize yourself with the main components of a console application and the looping structure.



Key Concepts

- 1 **`while` loop:** Repeats a block of code a fixed number of times (3 in this case).
- 2 **Counter Pattern:** ``count`` variable tracks the number of iterations.
- 3 **Accumulator Pattern:** ``sum``, ``totalEven``, ``totalOdd`` variables accumulate totals.



Program Structure

- 1 The main logic resides in the ``Main`` method of ``Program.cs``.
- 2 Initializes ``sum``, ``totalEven``, ``totalOdd``, and ``count`` to 0.
- 3 A ``while`` loop runs 3 times.

4

Conditional Statements (`if-else`): Determine if a number is even or odd using the modulo operator (`%`).

5

`int.Parse()`: Directly converts string input to an integer.

4

Inside the loop, it prompts for a number, adds it to `sum`, and increments `totalEven` or `totalOdd` based on parity.

5

After the loop, it displays the final `sum`, `totalEven`, and `totalOdd`.



Pro Tip: Direct parsing of user input can lead to crashes if the input is not in the expected format. In real-world applications, robust input validation (e.g., using `TryParse`) is essential.

Simulate

C# Console Simulation

Enter numbers and see their sum, and counts of even/odd numbers.

Simulated Console:

Enter a number:

e.g., 10

Enter Number

```
Console.Write("Enter a number: ");
```

Reset Simulation

Step 3

Write the C# Code

Implementing the logic for number analysis in `Program.cs`.



Code Implementation

- 1 Open the 'Program.cs' file in your project.
- 2 Replace the default 'Main' method content with the C# code below.
- 3 Pay close attention to the 'while' loop, accumulator variables, and the 'if-else' for even/odd checking.



Test Your Application

- 1 Press **F5** or click the **Start button** (green play icon) to run your application.
- 2 A console window will open.
- 3 Enter three numbers as prompted.
- 4 Observe the final sum, total even, and total odd numbers.

```
// In Program.cs
using System;

public class Prac12Question1
{
    public static void Main(string[] args)
    {
        int sum = 0;
        int totalEven = 0;
        int totalOdd = 0;
        int count = 0;

        while (count < 3)
        {
            Console.Write("Enter a number: ");
            // WARNING: This will crash if input is not a valid integer.
            int number = int.Parse(Console.ReadLine());

            sum += number;

            if (number % 2 == 0)
            {
                totalEven++;
            }
            else
            {
                totalOdd++;
            }
            count++;
        }

        Console.WriteLine("Sum: {0}", sum);
        Console.WriteLine("Total Even: {0}", totalEven);
        Console.WriteLine("Total Odd: {0}", totalOdd);

        Console.ReadLine();
    }
}
```

C#

🎉 You Did It! Your Number Analyzer is Ready! 🎉

✅ Project Created ✅ Logic Understood ✅ Code Implemented

Troubleshooting Common Issues

❌ Console Closes Immediately

The console window flashes and closes without waiting for output.

- ✓ Ensure `Console.ReadLine();` is the very last line in your `Main` method before the closing brace.
- ✓ This line pauses the console until you press Enter.

```
Add Console.ReadLine(); at the end of Main().
```

❌ Program Crashes on Input

The program terminates unexpectedly when entering a non-integer value.

- ✓ This program uses direct `int.Parse()`. If you enter text that is not a valid integer (e.g., "abc" or "10.5"), the program will crash.
- ✓ Ensure you enter only whole numbers as prompted.

```
Always enter valid integers. For robust applications, use
```

❌ Incorrect Sum or Counts

The calculated sum, total even, or total odd numbers are incorrect.

- ✓ Verify that `sum`, `totalEven`, and `totalOdd` are initialized to `0`.
- ✓ Ensure `sum += number;` is correctly placed inside the loop.
- ✓ Check the `if (number % 2 == 0)` condition and the corresponding `totalEven++` and `totalOdd++` increments.
- ✓ Confirm `count++` is inside the loop to ensure it runs exactly 3 times.

```
Step through the loop with the debugger (F10) to
```

❌ Loop Doesn't Run 3 Times

The program asks for more or fewer than 3 numbers.

- ✓ Ensure the `while` loop condition is `count < 3`.
- ✓ Make sure `count++` is correctly placed at the end of the loop body.

```
', and 'count'
```

